INTRODUCTION TO DATA STRUCTURE S

MIT SCHOOL OF COMPUTING

AS PART OF CURRICULAM

FIRST YEAR (SEM-2)

COMPUTER SCIENCEAND ENGINEERING

UNDER THE GUIDANCE OF

Prof. Sachin Tiwari

FOR THE ACADEMIC YEAR 2025-26

SUBMITTED BY

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EXPERIMENT DETAILS

Experiment Number	Experiment Name
	mg e e 1st
2	Doubly Linked List
3	

Virtual Lab NameVirtual Labs

:

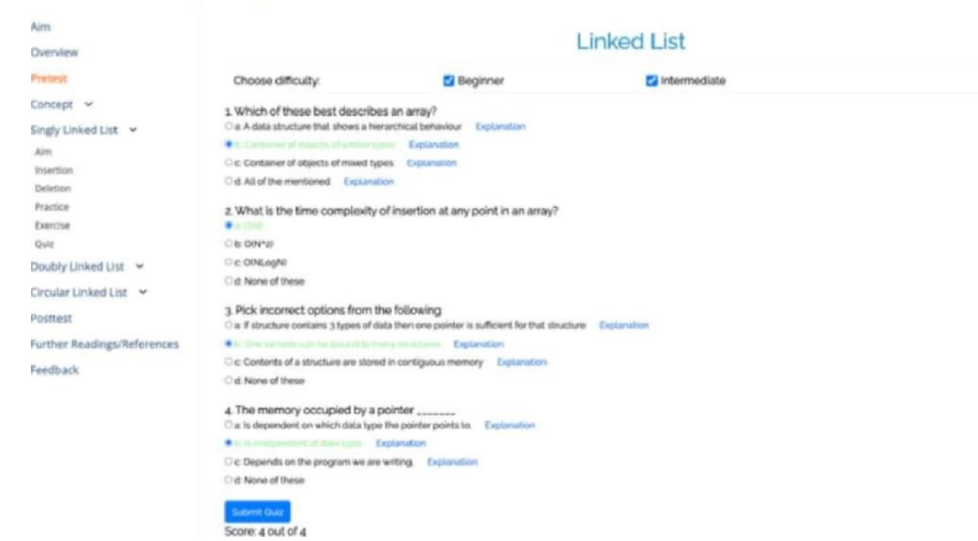
Institute:

111T-H(Hyderabad)

Date of Performance: 28 M arch 2026

SCREENSHOTS

Computer Soence and Engtneenng > Data Structures • > Expenrnents



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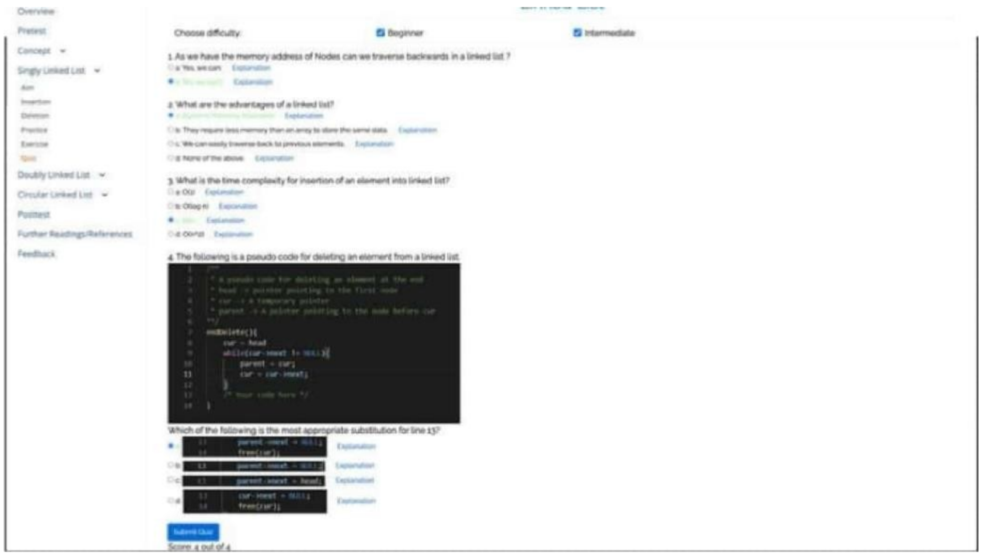
Linked List



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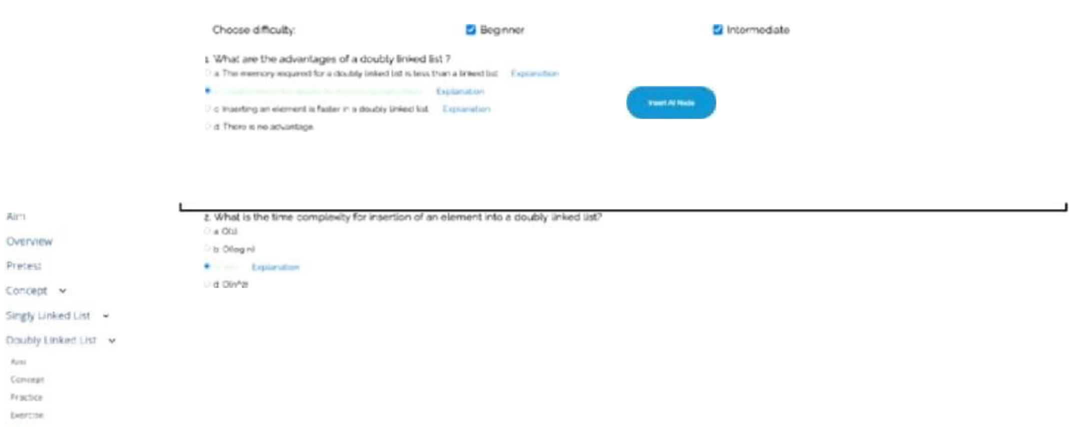

Linked List

posttest


Readings/References
3. Which of the following can make deletion in doubly linked list a constant time operation?
a. Keeping the doubly linked list sorted
Explanation
b. Having a pointer at both the first and last element
Explanation
Explanation
d. Knowing the position of the node
Explanation
Submit Quiz
Score 3 out of 3

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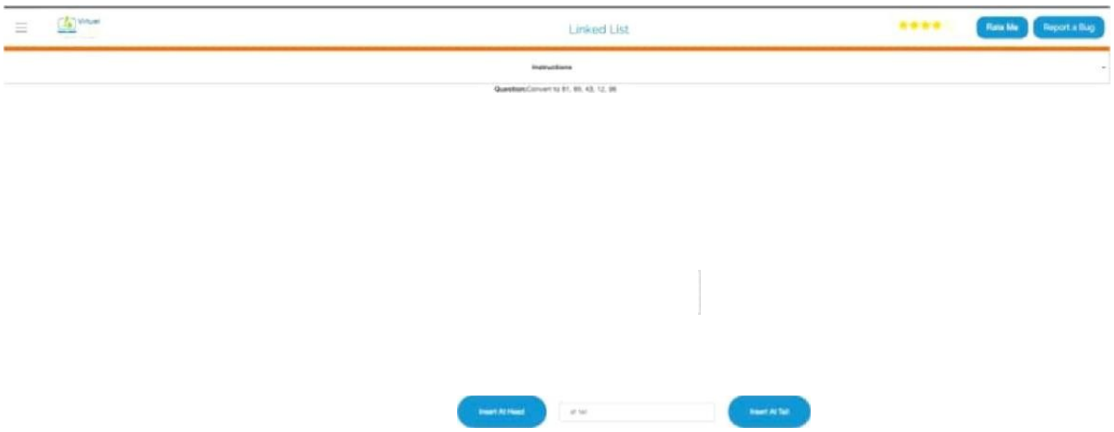
Linked List
Rate Me
Report a Bug
Instructions
Question: Convert to 81, 95, 43, 12, 98
Insert At Head
Insert At Tail

Virtual Labs

Rate Me Report a Bug

mputer Science and Engineering > Data Structures - 1 > Experiments

Linked List

- im
- Overview
- Pretest
- Concept
- Singly Linked List
- Doubly Linked List
- Circular Linked List
  - Aim
  - Concept
  - Practice
  - Exercise
  - Quiz
- Posttest
- Further Readings/References
- Feedback

1. In a circular linked list
- a. We can navigate through the linked list in reverse order
- c. Searching is faster than in a singly Linked list
- d. None of these

2. In a circular linked list, if we insert an element how many pointers do we have to update ?
- a. 0
- b. 1
- Explanation
- d. 3

3. What is the time complexity for deleting an element from circular linked list?
- a. O(1) Explanation
- b. O(log n)
- d. O(n^2)

Submit Quiz

Score: 3 out of 3

- Overview
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Choose difficulty: ☑ Beginner ☑ Intermediate

1. Pick one correct statement from the following?
- a. In a doubly linked list, each node is linked to one node
- b. In a doubly linked list, each node is linked to three nodes
- Explanation
- d. In a doubly linked list, each node is linked to zero nodes

2. What is the time complexity of searching for an element in a circular linked list?
- Explanation
- b. O(nlogn)
- c. O(1)
- d. None of the mentioned

3. Assuming memory is not a limitation which of the following linked list is the best for searching
- a. Singly Linked List
- Explanation
- c. Circular Linked List
- d. All give same performance

4. Consider the following linked list: 2 -> 4 -> 7 -> 9 -> 10 -> 13. Their addresses are 2408, 2560, 1560, 5440, 3240, 3000. Head variable is pointing at 4 (the second node). What will be head -> next -> next -> value.
(Node has value and next, value stores the value and next stores the address of the next variable)
- a. 7
- b. 10 Explanation
- d. 13

5. The first node of a linked list is at the location 2000. At which memory location will the second node be added to. Assume the nodes contain integers, each integer occupies a memory of 4 bits. And each pointer is 1 bit.
- a. 2004
- b. 2006
- c. 1995
- Explanation

Submit Quiz

Score: 5 out of 5

STUDENT DECLARATION

I here by declare that the Virtual Lab experiments included in this document have been performed by me as part of my academic activity. The screenshots submitted are genuine and taken during the actual execution of the experiments.

Student Signature: Aditya Jain

Date:26/04/2026